

Operations and Supply Management

Data Analysis Using Python

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1 Introduction

Operations and Supply Management involves planning, controlling, and optimizing the flow of materials, information, and products from suppliers to customers. Python provides powerful tools for analyzing supply chain data, evaluating supplier performance, monitoring inventory, and supporting decision making.

2 Objectives

The objectives of this analysis are:

- Clean operational data.
- Calculate important Key Performance Indicators (KPIs).
- Evaluate supplier performance.
- Analyze inventory status.
- Analyze product sales.
- Produce graphical reports.
- Export summarized reports.

3 Required Python Libraries

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
```

4 Input Dataset

The analysis assumes a CSV file named

`supply_chain.csv`

containing the following variables.

5 Data Cleaning

Typical preprocessing includes

- Removing duplicate records.
- Converting dates.
- Calculating lead time.
- Handling missing values.

Variable	Description
$Order_{ID}$	Order identification
$Order_{Date}$	Date of order
$Delivery_{Date}$	Date delivered
Supplier	Supplier name
Product	Product name
Quantity	Units ordered
$Unit_{Cost}$	Cost per unit
Revenue	Total revenue
$Average_{Inventory}$	Average inventory level
$Current_{stock}$	Current stock
$Reorder_{Level}$	Reorder threshold
Warehouse	Warehouse location
Customer	Customer name
Region	Sales region

6 Key Performance Indicators

The following KPIs are calculated.

- Total Orders
- Total Units Sold
- Total Revenue
- Average Lead Time
- Average Unit Cost

7 Supplier Performance

Supplier performance is summarized using

- Average lead time
- Total quantity supplied
- Total revenue generated

8 Inventory Analysis

Inventory turnover is computed as

$$InventoryTurnover = \frac{QuantitySold}{AverageInventory}$$

Stock status is classified as

- Reorder
- Sufficient

depending on the current stock level.

9 Monthly Sales Analysis

Monthly revenue is obtained by grouping orders according to month and summing revenue values.

10 Visualization

Python generates several useful graphs including

- Average Supplier Lead Time
- Monthly Revenue Trend
- Top Products by Revenue

These plots assist managers in identifying trends and improving operational performance.

11 Output Files

The analysis exports

- `supplier_performance.csv`
- `product_summary.csv`
- `monthly_sales.csv`

which can be imported into SAS, Excel, R, or other statistical software.

12 Conclusion

Python provides a comprehensive platform for Operations and Supply Management analytics. By integrating data cleaning, KPI calculation, inventory management, supplier evaluation, visualization, and reporting, organizations can improve operational efficiency, reduce costs, and support evidence-based decision making. The framework can be further extended with demand forecasting, machine learning, Monte Carlo simulation, and optimization algorithms.